Class 10 Mathematics – Important Practice Questions

Midterm Preparation | CBSE-style practice

1. Real Numbers

1. Use Euclid’s division algorithm to find the HCF of 867 and 255.

2. Find the HCF and LCM of 96 and 404 by prime factorisation.

3. Prove that $\sqrt{5}$ is irrational.

4. Prove that $3 + 2\sqrt{5}$ is irrational.

5. Determine whether $13/3125$ has a terminating or non-terminating decimal expansion.

6. Find the greatest number which divides 245 and 1029 leaving remainders 5 and 9 respectively.

7. The HCF of two numbers is 12 and their LCM is 420. If one number is 60, find the other.

8. Show that the square of any positive integer is of the form $3m$ or $3m + 1$.

2. Polynomials

1. Find the zeroes of $x^2 - 7x + 12$ and verify the relationship between the zeroes and coefficients.

2. Find the zeroes of $2x^2 - 5x - 3$.

3. If the zeroes of $x^2 - (k + 2)x + 2k$ are equal, find k.

4. Find a quadratic polynomial whose zeroes are 3 and –5.

5. One zero of $2x^2 + px + 6$ is 2. Find p and the other zero.

6. If α and β are the zeroes of $3x^2 - 7x + 2$, find $\alpha^2 + \beta^2$.

7. Find the zeroes of a polynomial from its given graph.

8. Divide $2x^3 + 3x^2 - 5x + 6$ by $x + 2$ and verify the division algorithm.

3. Pair of Linear Equations in Two Variables

1. Solve: 2x + 3y = 13 and x – y = 1.

2. Solve: 3x + 4y = 10 and 5x – 2y = 2.

3. Solve graphically: x + y = 5 and x – y = 1.

4. Find k for which 2x + 3y = 7 and (k + 1)x + (k + 2)y = 14 have infinitely many solutions.

5. Determine the nature of solutions of 3x + 2y = 5 and 6x + 4y = 12.

6. The sum of two numbers is 27 and their difference is 5. Find the numbers.

7. A two-digit number is four times the sum of its digits. If 18 is added to it, the digits are reversed. Find the number.

8. 2 pens and 3 pencils cost ■26, while 3 pens and 2 pencils cost ■29. Find the cost of each.

4. Quadratic Equations

1. Solve $x^2 - 7x + 12 = 0$.

2. Solve $2x^2 - 5x - 3 = 0$ using the quadratic formula.

3. Find the nature of the roots of $3x^2 - 4x + 5 = 0$.

4. Find k if $kx^2 - 6x + 9 = 0$ has equal roots.

5. Find k if $x^2 - 5x + k = 0$ has equal roots.

6. Solve $x^2 + 6x - 7 = 0$ by completing the square.

7. The product of two consecutive positive integers is 306. Find the integers.

8. The area of a rectangle is 528 m^2. Its length is 10 m more than its breadth. Find its dimensions.

5. Arithmetic Progressions

1. Find the 20th term of the AP 3, 7, 11, 15, …

2. Find the 25th term of the AP 18, 15, 12, 9, …

3. Which term of the AP 5, 9, 13, 17, … is 121?

4. Find the sum of the first 30 terms of 7, 10, 13, 16, …

5. Find the sum of the first 20 odd positive integers.

6. The 7th term of an AP is 24 and the 15th term is 56. Find the first term and common difference.

7. How many terms of the AP 8, 11, 14, … are required to get a sum of 550?

8. A student saves ■500 in the first month and increases the saving by ■50 every month. Find the total saving in 12 months.

6. Coordinate Geometry

1. Find the distance between (2, 3) and (8, 11).

2. Find the midpoint of (–3, 5) and (7, –1).

3. Find the point which divides (2, –3) and (8, 9) internally in the ratio 1:2.

4. Find the point which divides (–2, 3) and (4, –6) internally in the ratio 2:1.

5. Show that (1, 2), (4, 6) and (7, 10) are collinear.

6. Find the area of the triangle whose vertices are (2, 3), (–1, 5) and (4, –2).

7. Find k if (2, 3), (4, k) and (6, 7) are collinear.

8. In what ratio does (3, 4) divide the line segment joining (1, 2) and (7, 8)?

7. Areas Related to Circles

1. Find the area and circumference of a circle of radius 14 cm.

2. Find the area of a sector of radius 21 cm and angle 60°.

3. Find the length of the arc of a circle of radius 14 cm subtending 90° at the centre.

4. Find the area of the minor segment of a circle of radius 14 cm with central angle 90°.

5. A square has a circle inscribed in it. Find the area of the region outside the circle but inside the square.

6. A circular track has radius 35 m. Find the distance covered in 10 rounds.

7. Find the area of a quadrant of radius 14 cm.

8. A circle of radius 14 cm has a 90° sector removed. Find the area of the remaining part.

8. Triangles

1. State and prove the Basic Proportionality Theorem.

2. Two similar triangles have corresponding sides in the ratio 3:5. Find the ratio of their areas.

3. Prove the AA criterion for similarity of triangles.

4. Prove the SAS criterion for similarity of triangles.

5. In triangle ABC, DE ■ BC. If AD = 3 cm, DB = 2 cm and AE = 4.5 cm, find EC.

6. Prove the Pythagoras theorem.

7. A right triangle has hypotenuse 13 cm and one side 5 cm. Find the other side.

8. Prove the converse of the Pythagoras theorem.